

Daniel Daza

✉ dfdazac@gmail.com

🏠 dfdazac.github.io

🎓 [Google Scholar](#)

[in](#) [LinkedIn](#)

Summary	My research focuses on artificial intelligence methods for representing and learning from structured scientific data such as graphs. This includes methods for constructing structured representations, such as building and maintaining knowledge graphs from heterogeneous and textual data using language models, and methods for learning from structured representations, including foundation models that combine knowledge graphs with multimodal scientific data to support prediction, reasoning, and scientific discovery.	
Education	Vrije Universiteit Amsterdam	September 2019 - June 2024
	<i>Doctor of Philosophy, cum laude</i>	Amsterdam, The Netherlands
	Thesis: Exploiting Subgraphs and Attributes for Representation Learning on Knowledge Graphs Supervisors: prof. Frank van Harmelen (Vrije Universiteit Amsterdam), prof. Paul Groth (University of Amsterdam), Dr. Michael Cochez (Vrije Universiteit Amsterdam).	
	University of Amsterdam	September 2017 - August 2019
	<i>MSc in Artificial Intelligence, cum laude</i>	Amsterdam, The Netherlands
	Thesis: A Modular Framework for Unsupervised Graph Representation Learning Supervisor: Dr. Thomas Kipf	
	UD Francisco José de Caldas	August 2010 - August 2016
	<i>BSc in Electronics Engineering</i>	Bogotá, Colombia
	Thesis: Neural Networks for Learning and Optimizing Electrical Machine Surrogates.	
Current employment	Eindhoven University of Technology	September 2026 - Present
	<i>Postdoctoral Researcher</i>	Eindhoven, Netherlands
	My research within the Horizon Europe SimuLingua project focuses on knowledge graphs and multimodal foundation models for materials discovery, connecting scientific literature, materials data, and simulations.	
Work experience	Vrije Universiteit Amsterdam	January 2026 - August 2026
	<i>Postdoctoral Researcher</i>	Amsterdam, Netherlands
	I carried out research on foundation models for knowledge graphs, simulation-based model validation, and applications to drug-induced kidney disease.	
	Amsterdam UMC	July 2024 - December 2025
	<i>Postdoctoral Researcher</i>	Amsterdam, Netherlands
	In collaboration with Accenture and clinicians, I carried out research in the field of machine learning for knowledge graphs, and its applications to rare diseases in the healthcare domain.	
	Vrije Universiteit Amsterdam	November 2019 - June 2024
	<i>PhD Researcher</i>	Amsterdam, Netherlands
	I carried out research in representation learning for knowledge graphs, and in relation to the same topic I supervised bachelor and master students, and contributed to course organization and lectures.	

Bosch Research Center for Artificial Intelligence*Research Intern*

May 2023 - August 2023

Renningen, Germany

Industrial research internship on the topics of unsupervised learning on graphs for explainable similarity search.

Irdeto B.V.*Data Science Intern*

June 2018 - August 2018

Hoofddorp, The Netherlands

Industrial internship on the development and deployment of scalable machine learning models with Tensorflow on Kubernetes clusters and cloud storage, applied to the problem of fraud detection.

Teaching*Machine Learning for Graphs* (MSc), Vrije Universiteit Amsterdam

January 2026

Contributed two lectures on “Deep Graph Representation Learning” and “Complex Query Answering.”

Deep Learning (MSc), Vrije Universiteit Amsterdam

November 2024

Contributed one lecture on “Learning with Graphs.”

Machine Learning for Graphs (MSc), Vrije Universiteit Amsterdam

January 2023

Contributed to the design of the first edition of this course, now a recurring MSc course at VU Amsterdam; contributed one lecture on “Approximate Query Answering over Knowledge Graphs”; designed one homework assignment; guided student working sessions; and graded homework.

AI in Practice: Preparing for AI (MOOC), TU Delft

Invited during my PhD to contribute a talk on machine learning for knowledge graphs.

Supervision**Master Students**

Paul Constantinescu, “*Zero-shot Entity Disambiguation with Language Models and Search Indexes*”, (MSc thesis, Vrije Universiteit Amsterdam, ongoing).

Akshaye Iyer, “*Neuro-Symbolic Knowledge Graph Construction using Natural Language Understanding*”, (MSc thesis, Vrije Universiteit Amsterdam, ongoing).

Baradwaj Varadharajan, “*Inductive Link Prediction over Novel Relations*” (MSc thesis, University of Amsterdam, 2023).

Qingzhi Hu, “*Data Integration and Predictive Modeling for Impact Investing*” (MSc thesis, University of Amsterdam, 2022).

Stefan Schouten, “*Incorporating Semantics in Knowledge Graph Embeddings*” (MSc thesis, University of Amsterdam, 2021), with Thiviyan Thanapalasingam.

Bachelor Students

Vencel Kuba, “*Evaluating Analogical Reasoning in Knowledge Graph Foundation Models*”, (BSc thesis, Vrije Universiteit Amsterdam, ongoing with Hossein Khojasteh).

Kate Jermakova, “*Structure-Aware Query Corruption in Neural Knowledge Graph Reasoning*”, (BSc thesis, Vrije Universiteit Amsterdam, 2025).

Sławek Męczyński, “*Enhancing Link Prediction in Knowledge Graphs Through Pre-Informed Training*” (BSc thesis, Vrije Universiteit Amsterdam, 2025).

Fredrik Skjeltvik, “*Complex Query Answering in the Biomedical Domain*” (BSc thesis, Vrije Universiteit Amsterdam, 2022).

Communication and Outreach	AI & Mathematics Network , Tilburg, The Netherlands Learning on Knowledge Graphs for Scientific Discovery	June 2025
	ELLIS workshop on Representation Learning and Generative Models for Structured Data , Amsterdam, The Netherlands Interactive Query Answering on Knowledge Graphs with Soft Entity Constraints	February 2025
	Austrian Institute of Technology , Vienna, Austria Learning on Graphs via Multimodal Data	February 2024
	Deloitte , Amsterdam, The Netherlands Learning Entity Representations from Knowledge Graphs and Textual Descriptions	June 2022
	Zeta Alpha , Amsterdam, The Netherlands Inductive Entity Representations from Text via Link Prediction	September 2021
	King’s College London , London, UK Complex Query Answering with Neural Link Predictors	March 2021
	Elsevier , Amsterdam, The Netherlands Message Passing Query Embedding	February 2020
Grants and Distinctions	Public communication ICT.OPEN video: <i>Discovering Missing Links in Knowledge</i> , public-facing video about research in our lab https://youtu.be/mF94YDg4KPE	2022
	Gold Reviewer	ICML, 2026
	Young Researcher and Innovator Conference Grant Awarded €1500 for participation in the <i>International Conference on Artificial Intelligence in Medicine (AIME 2026)</i> .	COST Action CA23147, 2026
	Computing Time on National Computing Facilities (lead co-applicant) Project: <i>Large Laboratory Models</i> , awarded 300k CPU+2M GPU compute credits.	NWO, 2025
	Doctorate cum laude (top 5%)	Vrije Universiteit Amsterdam, 2024
	Best Paper Honorable Mention	Learning on Graphs Conference, 2024
	Outstanding Paper Award	ICLR, 2021
Academic Service	Reviewing Transactions on Machine Learning Research (TMLR)	2026
	International Conference on Learning Representations (ICLR)	2025, 2026
	Learning on Graphs Conference	2025, 2026
	Advances in Neural Information Processing Systems (NeurIPS)	2025, 2026
	International Conference on Machine Learning (ICML)	2025, 2026
	The Web Conference	2024, 2025, 2026

ACM Transactions on Knowledge Discovery and Data	2023
ACL Workshop on Structured Predictions for NLP	2022
Conference on Information and Knowledge Management (CIKM)	2022
Semantic Web Journal	2020, 2025, 2026

Tutorials

Reasoning beyond Triples: Recent Advances in Knowledge Graph Embeddings (CIKM 2023).

Publications

*Indicates equal contribution.

2026

Select, Don't Train: The Benefits of Modular Entity Disambiguation with LLM-Based Selection, in **ISWC 2026**.

F. Polat, D. Daza, P. Zhang, K. Zaporozhets and P. Groth.

Interactive Query Answering on Knowledge Graphs with Soft Entity Constraints, in **TMLR**.
D. Daza, A. Bernardi, L. Costabello, C. Gueret, M. Mansoury, M. Cochez, M. Schut.

Contextualized Interpretation of Machine-Learning Predictions in Rare Disease Omics: Integrating SHAP, Biomedical Knowledge, and Language Models, in **International Conference on Artificial Intelligence in Medicine**.

D. Daza, Y. Jaspers, A. Bernardi, L. Costabello, C. Gueret, M. Cochez, M. Engelen, S. Kemp, M. Schut.

Explaining Graph Neural Networks for Node Similarity on Graphs, in **TMLR**.

D. Daza, C.X. Chu, T.K. Tran, D. Stepanova, M. Cochez, P. Groth.

Counting Still Counts: Understanding Neural Complex Query Answering Through Query Relaxation, in **TMLR**.

Y. Brunink*, D. Daza*, Y. He, M. Cochez.

Similarity-Constrained Reweighting for Complex Query Answering on Knowledge Graphs, **Under review**.

D. Daza, A. Bernardi, L. Costabello, C. Gueret, M. Cochez, M. Schut.

2025

Discovering Association Rules in High-Dimensional Small Tabular Data, in **ECAI 2025 Workshop on Advanced Neuro-Symbolic Applications**.

E. Karabulut, D. Daza, P. Groth, V. Degeler.

EMERGE: A Benchmark for Updating Knowledge Graphs with Emerging Textual Knowledge, **Under review**.

K. Zaporozhets, D. Daza, E. Barba, I. Assent, R. Navigli, P. Groth.

GRAPES: Learning to sample graphs for scalable graph neural networks, in **TMLR**.

T. Younesian, D. Daza, E. van Krieken, T. Thanapalasingam, P. Bloem.

2024

UnRavL: A Neuro-Symbolic Framework for Answering Graph Pattern Queries in Knowledge Graphs, in **Learning on Graphs**.

🏆 Honorable Mention for Best Paper

T. Cucumides, D. Daza, P. Barcelo, M. Cochez, F. Geerts, J.L. Reutter, M.R. Orth.

2023

BioBLP: a modular framework for learning on multimodal biomedical knowledge graphs, in **Journal of Biomedical Semantics**.

D. Daza, D. Alivanistos, P. Mitra, T. Pijnenburg, M. Cochez, P. Groth.

Adapting Neural Link Predictors for Data-Efficient Complex Query Answering, in **NeurIPS**.

E. Arakelyan, P. Minervini, D. Daza, M. Cochez, Isabelle Augenstein.

Harnessing the Web and Knowledge Graphs for Automated Impact Investing Scoring, in **KDD Workshop on AI for Climate Sustainability**.

Q. Hu, D. Daza, L. Swinkels, K. Üsaitė, R. Hoen, and P. Groth .

2022

SlotGAN: Detecting Mentions in Text via Adversarial Distant Learning, in **ACL Workshop on Structured Prediction for NLP**.

D. Daza, M. Cochez, and P. Groth.

2021

Complex Query Answering with Neural Link Predictors, in **ICLR**.

🏆 Outstanding Paper Award (top 1%)

E. Arakelyan*, D. Daza*, P. Minervini*, and M. Cochez.

Entity Representations from Text via Link Prediction, in **The Web Conference**.

D. Daza, M. Cochez, and P. Groth.

Approximate knowledge graph query answering: from ranking to binary classification, in **ECAI 2020 Workshop on Graphs for Knowledge Representation and Reasoning**.

R. van Bakel, T. Aleksiev, D. Daza, D. Alivanistos, and M. Cochez.

2020

Message passing query embedding, in **ICML 2020 Workshop on Graph Representation Learning**.

D. Daza and M. Cochez.